πthons

HRT Challenge for the 2026 Datathon

We predict once at bar 50: first stay out of bad tails, then size the trades worth taking.

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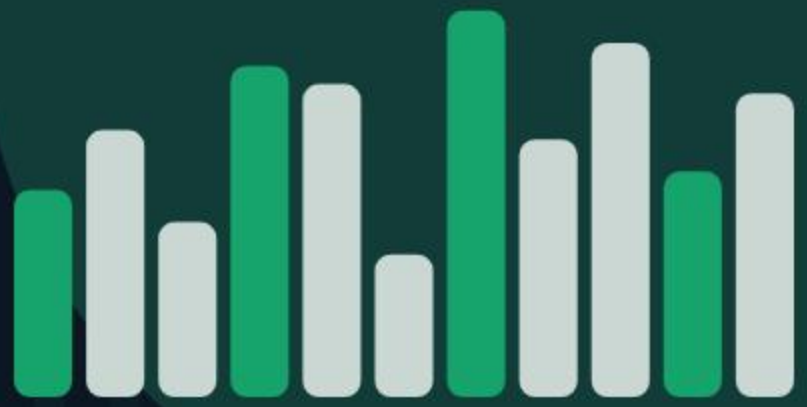
Submission model: subspace-btp-hdoc-ensemble



Core thesis

Stay out of bad tails. Size the trades worth taking.

Position once, win on Sharpe

At bar 50 we choose one target position for the rest of the session. Sharpe rewards a smooth PnL path, so sizing matters as much as direction.

Seen half

OHLC bars + headlines

Position

target_position

Held to close

PnL = position × future return

What this means

We would rather skip a noisy setup than carry full size into a bad tail.

1,000	20,000	50	Sharpe
training sessions	test sessions	seen bars per session	evaluation metric

Signals we trusted

Every feature is built from the seen half. Price movement gives the base view; headlines add event context and timing.

FEATURES

- 426 price-path features
- bar-50 cutoff only
- same schema for public/private

Seen bars

OHLC path up to bar 50

Price path

returns, ranges, candle body, DCT shape

Headlines

templates, intents, transitions, timing

Risk view

bad-tail odds + downside rank

Position

calibrated target_position

Headline-aware downside filter

Headlines matter most when they tell us a session is unusual, late-breaking, or moving into a risky template.

HEADLINE SHAPE

- templates and intent
- sequence changes
- late headlines

DOWNSIDE GATE

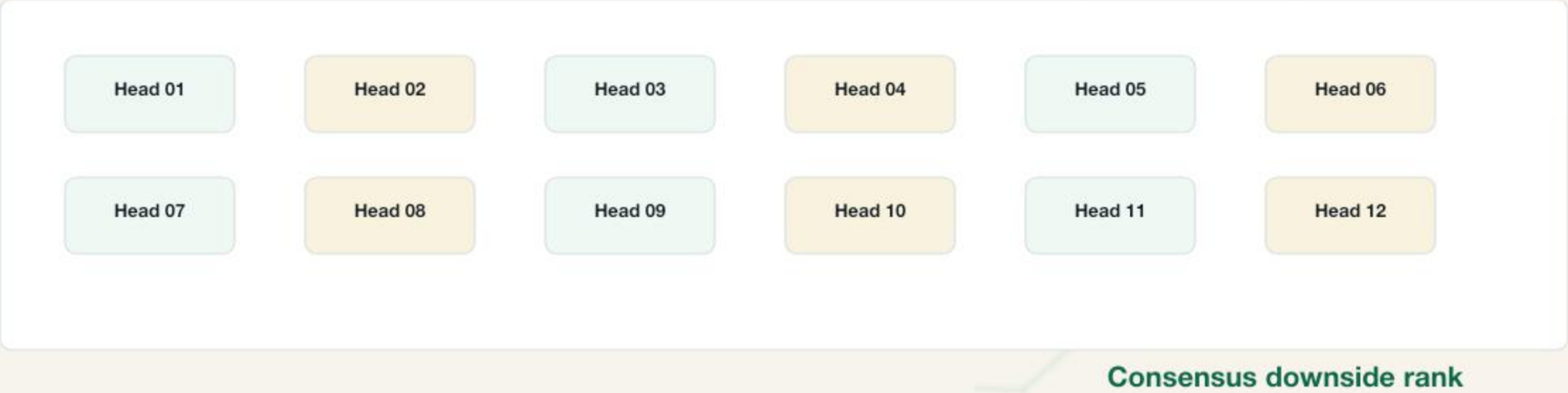
- ExtraTrees classifier
- bad-tail focus
- loss avoidance

SIZING

- trim flagged sessions
- scale remaining exposure
- keep long bias controlled

Subspace bagged downside ranker

Twelve logistic heads each see a different slice of the price feature set. We trust heads that validate well and let consensus drive the rank.



Controlled 70/30 ensemble

The subspace ranker sets the exposure shape. The headline model can move sessions within that shape when text evidence disagrees.

70%

price anchor

30%

headline check

Price anchor

subspace downside ranker

Headline check

document/template ranker

70 / 30

level_qmap

Submission

one position per test session

What we submit

The final file is a 20k-row position vector: broad enough to take opportunities, cautious enough to step away from the worst tails.

20,000
target positions

3.79
local train Sharpe

12
price heads

70/30
price/headline mix

PROTECT SHARPE

- gate bad tails first
- do not chase every signal

USE BOTH VIEWS

- price path plus headlines
- agreement raises conviction

KEEP SIZE SANE

- rank/quantile mapping
- controlled exposure spread